IBFE MATHEMATICAL FORMULATION

Strong form, weak form, and discrete coupling

Laura A. Miller

Department of Mathematics, University of Arizona

Fluid Dynamics and IBM Theory • Module 4

$$W(\boldsymbol{F}) \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{\mathcal{F}}_h \longrightarrow \boldsymbol{f}_h$$

Solid mechanics in reference coordinates, fluid dynamics on a Cartesian grid.

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

ROADMAP FROM STRESS TO COUPLED MOTION

1. Describe deformation by the motion $\chi(\boldsymbol{X}, t)$ and its gradient $\boldsymbol{F}$.
2. Convert a constitutive stress $\boldsymbol{P}$ into a Lagrangian force by virtual work.
3. Spread that force to the Eulerian fluid grid with a regularized delta kernel.
4. Restrict fluid velocity back to the finite-element structure with an adjoint operator.

> The finite-element mesh represents geometry and mechanics. Quadrature points mediate the fluid–structure interaction.

The formulation in this deck (strong form, unified and partitioned weak forms, quadrature coupling, adjoint restriction) is adapted from Griffith & Luo (2017), which builds on Boffi, Gastaldi, Heltai & Peskin (2008); the stabilization remarks follow Griffith & Patankar (2020) and the jellyfish example Hoover, Griffith & Miller (2017). Equation numbers on individual slides refer to Griffith & Luo. Full references are on the final slide.

NOTATION BRIDGE FROM THE SOLID-MECHANICS MODULE

Kinematics and stress

$$\boldsymbol{F} = \nabla_X \chi, \qquad J = \det \boldsymbol{F}, \qquad \boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}}.$$

- $\boldsymbol{F}$: deformation gradient;
- $\boldsymbol{P}$: first Piola–Kirchhoff stress;
- W: energy per reference volume.

Force and velocity coupling

- $\boldsymbol{\mathcal{F}}(\boldsymbol{X}, t)$: Lagrangian force per reference volume;
- $\boldsymbol{f}(\boldsymbol{x}, t)$: Eulerian force per current volume;
- $\chi_t = \boldsymbol{u} \circ \chi$: no-slip motion.

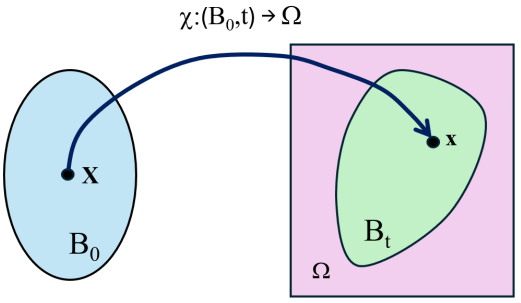
> Here we write the reference and current solid regions as B_0 and B_t, reserving Ω for the Eulerian computational domain.

REFERENCE AND CURRENT CONFIGURATIONS

$$\chi(\cdot, t) : B_0 \longrightarrow B_t \subset \Omega, \qquad \boldsymbol{x} = \chi(\boldsymbol{X}, t).$$

$$\boldsymbol{F}(\boldsymbol{X}, t) = \nabla_X \chi(\boldsymbol{X}, t),$$
$$J(\boldsymbol{X}, t) = \det \boldsymbol{F}(\boldsymbol{X}, t) > 0.$$

- $\boldsymbol{X} \in B_0$: material coordinate.
- $\boldsymbol{x} \in \Omega$: spatial coordinate.
- $B_t = \chi(B_0, t)$: region occupied by the immersed solid.



Schematic: instructor's course notes.

IBFE keeps the structural description Lagrangian while momentum, viscosity, and incompressibility remain Eulerian.

FLUID AND STRUCTURAL STRESS MEASURES

Newtonian fluid stress

$$\boldsymbol{\sigma}^f = -p\boldsymbol{I} + \mu\Big(\nabla_x \boldsymbol{u} + (\nabla_x \boldsymbol{u})^{\mathsf{T}}\Big).$$

Structural stress transformation

$$\boldsymbol{\sigma}^s(\chi(\boldsymbol{X},t),t) = \frac{1}{J}\boldsymbol{P}(\boldsymbol{X},t)\boldsymbol{F}^{\mathsf{T}},$$

$$\boldsymbol{P} \equiv \boldsymbol{P}^s = \boldsymbol{P}^p + \boldsymbol{P}^a, \qquad \boldsymbol{P}^p = \frac{\partial W}{\partial \boldsymbol{F}}.$$

Inside and outside the solid

$$\boldsymbol{\sigma}(\boldsymbol{x},t) = \begin{cases} \boldsymbol{\sigma}^f + \boldsymbol{\sigma}^s, & \boldsymbol{x} \in B_t, \\ \boldsymbol{\sigma}^f, & \boldsymbol{x} \in \Omega \setminus B_t. \end{cases}$$

The superscript s denotes the total structural stress. In a passive calculation, $\boldsymbol{P}^a = \boldsymbol{0}$.

> The Piola transform $\boldsymbol{P}^s = J\boldsymbol{\sigma}^s\boldsymbol{F}^{-\mathsf{T}}$ converts between current-area and reference-area stress measures.

CLASSICAL NEUTRALLY BUOYANT IBFE ASSUMPTIONS

Fluid and solid properties

- viscous incompressible fluid;
- neutrally buoyant immersed solid;
- one Eulerian density ρ;
- one fluid-like viscous stress $\boldsymbol{\sigma}^f$.

Coupling assumptions

- no slip and no penetration;
- the structure moves with $\boldsymbol{u}$;
- solid inertia is carried by the Eulerian momentum equation;
- there is no separate $\rho_0 \chi_{tt}$ equation in the coupled model.

> Extensions can treat density or viscosity contrasts, but they require additional terms beyond the classical formulation used here.

CONTINUOUS COUPLED EQUATIONS

$$\boxed{\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla_x)\boldsymbol{u}\right) = -\nabla_x p + \mu\nabla_x^2\boldsymbol{u} + \boldsymbol{f}}, \qquad \boxed{\nabla_x\cdot\boldsymbol{u} = 0}.$$

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_{B_0} \big(\nabla_X\cdot\boldsymbol{P} + \boldsymbol{G}\big)\delta(\boldsymbol{x}-\chi(\boldsymbol{X},t))\,d\boldsymbol{X} - \int_{\partial B_0} \boldsymbol{P}\boldsymbol{N}\,\delta(\boldsymbol{x}-\chi(\boldsymbol{X},t))\,dA.$$

$$\boxed{\frac{\partial\chi}{\partial t}(\boldsymbol{X},t) = \int_\Omega \boldsymbol{u}(\boldsymbol{x},t)\delta(\boldsymbol{x}-\chi(\boldsymbol{X},t))\,d\boldsymbol{x} = \boldsymbol{u}(\chi(\boldsymbol{X},t),t)}.$$

Here $\boldsymbol{G}$ is a prescribed force per reference volume. The continuum equations use the singular Dirac distribution δ; discretization replaces it by δ_h.

Griffith and Luo (2017), Eqs. (4)–(7), extended here by the body-force term $\boldsymbol{G}$.

VOLUME FORCE AND TRANSMISSION FORCE

$$\boldsymbol{f} = \underbrace{\int_{B_0} (\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G})\, \delta(\boldsymbol{x} - \chi)\, d\boldsymbol{X}}_{\text{volume contribution}} + \underbrace{\int_{\partial B_0} (-\boldsymbol{P}\boldsymbol{N})\, \delta(\boldsymbol{x} - \chi)\, dA}_{\text{transmission contribution}}.$$

Contribution	Reference support	Character in the fluid
$\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G}$	throughout B_0	regular body-force density
$-\boldsymbol{P}\boldsymbol{N}$	on ∂B_0	singular interfacial force layer

A nonzero transmission force can generate jumps in pressure or viscous stress. Omitting it from the strong form changes the model unless its effect is retained weakly.

FIRST PIOLA–KIRCHHOFF TRACTION

For a reference surface with unit normal $\boldsymbol{N}$,

$$\boldsymbol{t}_0 = \boldsymbol{P}\boldsymbol{N}, \qquad (t_0)_i = P_{iI} N_I.$$

$$\boldsymbol{P} = J\boldsymbol{\sigma}^s \boldsymbol{F}^{-\mathsf{T}}, \qquad \boldsymbol{\sigma}^s = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^{\mathsf{T}}.$$

$$[\boldsymbol{P}] = [\boldsymbol{t}_0] = \text{force/reference area}.$$

- i is a spatial index.
- I is a material index.
- $\boldsymbol{P}$ is generally nonsymmetric.

$\boldsymbol{P}\boldsymbol{N}$ is the traction on the solid. The transmission force $-\boldsymbol{P}\boldsymbol{N}$ is not a third-law reaction: it is the singular surface layer that appears because $\boldsymbol{\sigma}^s$ is truncated at ∂B_t (distributionally, $\nabla_x \cdot (\boldsymbol{\sigma}^s \chi_{B_t}) = (\nabla_x \cdot \boldsymbol{\sigma}^s)\chi_{B_t} - (\boldsymbol{\sigma}^s \boldsymbol{n})\,\delta_{\partial B_t}$, with χ_{B_t} the indicator function of B_t).

WHERE THE TRANSMISSION FORCE COMES FROM

Let χ_{B_t} be the indicator of the solid region (1 in B_t, 0 outside; not to be confused with the motion χ) and $\boldsymbol{n}$ the *outward* unit normal on ∂B_t. As distributions,

$$\nabla_x \chi_{B_t} = -\boldsymbol{n}\,\delta_{\partial B_t}, \qquad \partial_j\big(\sigma^s_{ij}\chi_{B_t}\big) = (\partial_j \sigma^s_{ij})\,\chi_{B_t} + \sigma^s_{ij}\,\partial_j \chi_{B_t} = (\partial_j \sigma^s_{ij})\,\chi_{B_t} - \sigma^s_{ij} n_j\,\delta_{\partial B_t}.$$

Eulerian reading
The structural stress contributes $\boldsymbol{f} = \nabla_x \cdot (\boldsymbol{\sigma}^s \chi_{B_t})$ to the momentum equation: a regular body force inside B_t plus a surface layer $-\boldsymbol{\sigma}^s\boldsymbol{n}$ on ∂B_t.

Lagrangian reading
Pulling back with $dv = J\,dV$, $\boldsymbol{n}\,da = J\boldsymbol{F}^{-\mathsf{T}}\boldsymbol{N}\,dA$, and $\boldsymbol{P} = J\boldsymbol{\sigma}^s\boldsymbol{F}^{-\mathsf{T}}$ turns the two pieces into $\nabla_X \cdot \boldsymbol{P}$ on B_0 and $-\boldsymbol{P}\boldsymbol{N}$ on ∂B_0: exactly the two integrals of the strong form.

> Sign check: the gradient of an indicator points *into* the region, hence the minus sign. The identity is the distributional form of the divergence theorem, $\int_{B_t} \nabla_x \cdot \boldsymbol{\sigma}^s\,dv = \int_{\partial B_t} \boldsymbol{\sigma}^s\boldsymbol{n}\,da$, which makes the total transmitted force $\int \boldsymbol{f}\,d\boldsymbol{x} = \boldsymbol{0}$ when no body force acts.

FORCE BALANCE ON A MATERIAL SUBREGION

Let $V_0 \subset B_0$ be fixed in reference coordinates. Its contact and body forces are

$$\int_{\partial V_0} \boldsymbol{P}\boldsymbol{N}\, dA, \qquad \int_{V_0} \boldsymbol{G}\, d\boldsymbol{X}.$$

The reference divergence theorem gives

$$\int_{\partial V_0} \boldsymbol{P}\boldsymbol{N}\, dA = \int_{V_0} \nabla_X \cdot \boldsymbol{P}\, d\boldsymbol{X}.$$

Hence the net force on V_0 is

$$\int_{V_0} (\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G})\, d\boldsymbol{X}.$$

> $\nabla_X \cdot \boldsymbol{P}$ is a spatial force vector per reference volume even though differentiation occurs with respect to material coordinates.

REFERENCE MOMENTUM BALANCE AND THE IBFE BRIDGE

For a stand-alone solid, localization of material momentum balance gives

$$\rho_0(\boldsymbol{X})\,\chi_{tt} = \nabla_X \cdot \boldsymbol{P} + \boldsymbol{G} \qquad \text{in } B_0.$$

Stand-alone solid
The Lagrangian equation carries solid inertia through $\rho_0 \chi_{tt}$.

Neutrally buoyant IBFE
The common Eulerian momentum equation carries fluid and solid inertia. The structural model supplies force, not a second inertia equation.

> IBFE spreads $\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G}$ and the boundary transmission force $-\boldsymbol{P}\boldsymbol{N}$ to the Eulerian grid. (It is the surface layer of the truncated structural stress, not a Newton's-third-law reaction; see "Where the transmission force comes from".)

WHY IBFE USES A WEAK STRUCTURAL FORCE

- C^0 finite elements give a piecewise stress, so $\nabla_X \cdot \boldsymbol{P}$ need not exist pointwise across element boundaries.
- Virtual work needs only first derivatives of the motion and the test function.
- The boundary transmission force remains part of the formulation.
- The result is a Lagrangian force field ready for Eulerian spreading.

> The weak form is the structural-force definition used by the numerical method, not merely an alternative notation for a smooth strong solution.

TEST FUNCTIONS AND DOUBLE CONTRACTION

Choose an arbitrary smooth Lagrangian test function $\boldsymbol{V}(\boldsymbol{X})$. For $\boldsymbol{P}, \nabla_X \boldsymbol{V} \in \mathbb{R}^{d\times d}$,

$$\boldsymbol{P} : \nabla_X \boldsymbol{V} = \operatorname{tr}\Big(\boldsymbol{P}^{\mathsf{T}} \nabla_X \boldsymbol{V}\Big) = \sum_{i=1}^{d}\sum_{I=1}^{d} P_{iI}\frac{\partial V_i}{\partial X_I}.$$

Multiplying the interior force by $\boldsymbol{V}$ and integrating gives

$$\int_{B_0} (\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G}) \cdot \boldsymbol{V}\, d\boldsymbol{X}.$$

> The contraction pairs a spatial force index i with a material derivative index I, consistent with the two-point character of $\boldsymbol{P}$.

INTEGRATION BY PARTS RETAINS THE BOUNDARY TERM

The tensor product rule is

$$\nabla_X \cdot (\boldsymbol{P}^{\mathsf{T}} \boldsymbol{V}) = (\nabla_X \cdot \boldsymbol{P}) \cdot \boldsymbol{V} + \boldsymbol{P} : \nabla_X \boldsymbol{V}.$$

After integration over B_0,

$$\int_{B_0} (\nabla_X \cdot \boldsymbol{P}) \cdot \boldsymbol{V} \, d\boldsymbol{X} = -\int_{B_0} \boldsymbol{P} : \nabla_X \boldsymbol{V} \, d\boldsymbol{X} + \int_{\partial B_0} (\boldsymbol{P}\boldsymbol{N}) \cdot \boldsymbol{V} \, dA.$$

> No traction-free assumption is made. The next two formulations account for the transmission force $-\boldsymbol{P}\boldsymbol{N}$ either implicitly in one weak density or explicitly as a surface density.

UNIFIED WEAK FORM

Define one Lagrangian force density $\boldsymbol{\mathcal{F}}$ by

$$\boxed{\int_{B_0} \boldsymbol{\mathcal{F}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = -\int_{B_0} \boldsymbol{P} : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} + \int_{B_0} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{X} \quad \text{for every } \boldsymbol{V}}.$$

The test functions are *not* required to vanish on ∂B_0. Integration by parts gives

$$\int_{B_0} \boldsymbol{\mathcal{F}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = \int_{B_0} (\nabla_X \cdot \boldsymbol{P} + \boldsymbol{G}) \cdot \boldsymbol{V}\, d\boldsymbol{X} - \int_{\partial B_0} (\boldsymbol{P}\boldsymbol{N}) \cdot \boldsymbol{V}\, dA.$$

> The single weak density $\boldsymbol{\mathcal{F}}$ packages the interior force and the boundary transmission force. Independently prescribed external tractions would still contribute separate boundary work.

Griffith and Luo (2017), Eqs. (10)–(14).

PARTITIONED WEAK FORM

Keep the volume and boundary forces separate. Find $\boldsymbol{\mathcal{F}}_{\text{vol}}$ such that

$$\int_{B_0} \boldsymbol{\mathcal{F}}_{\text{vol}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = -\int_{B_0} \boldsymbol{P} : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} + \int_{\partial B_0} (\boldsymbol{P}\boldsymbol{N}) \cdot \boldsymbol{V}\, dA + \int_{B_0} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{X}.$$

Then define

$$\boldsymbol{T} = -\boldsymbol{P}\boldsymbol{N} \quad \text{on } \partial B_0,$$

and spread the two fields independently:

$$\boldsymbol{f} = \int_{B_0} \boldsymbol{\mathcal{F}}_{\text{vol}}\, \delta(\boldsymbol{x} - \chi)\, d\boldsymbol{X} + \int_{\partial B_0} \boldsymbol{T}\, \delta(\boldsymbol{x} - \chi)\, dA.$$

Griffith and Luo (2017), Eqs. (15)–(21).

UNIFIED AND PARTITIONED FORMULATIONS

Unified

- one total Lagrangian density $\mathcal{F}$;
- boundary transmission represented through its weak definition;
- compact assembly and one spreading operator.

Partitioned

- volume density $\mathcal{F}_{\text{vol}}$ plus surface density $\boldsymbol{T}$;
- transmission force spread directly from ∂B_0;
- can improve accuracy for nonzero traction and a relatively coarse FE mesh.

> The formulations are equivalent in the continuum. Their finite-dimensional approximations differ, so neither discrete form is universally superior.

Griffith and Luo (2017), §6 and §7: with relatively coarse Lagrangian meshes the partitioned form gave more accurate pressures and better volume conservation, whereas only the unified form satisfies the discrete power identity (which may be needed for unconditionally stable implicit time stepping). When $\boldsymbol{PN} \equiv \boldsymbol{0}$ (e.g. their anisotropic shell, §6.1.1) the two forms coincide.

COMPLETE CONTINUOUS COUPLING CYCLE

1. Evaluate the structural stress $\boldsymbol{P}(\boldsymbol{F})$.
2. Define $\boldsymbol{\mathcal{F}}$ by virtual work:

$$\int_{B_0} \boldsymbol{\mathcal{F}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = -\int_{B_0} \boldsymbol{P} : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} + \int_{B_0} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{X}.$$

3. Spread force and solve the incompressible fluid equations:

$$\boldsymbol{f}(\boldsymbol{x}, t) = \int_{B_0} \boldsymbol{\mathcal{F}}(\boldsymbol{X}, t)\, \delta(\boldsymbol{x} - \chi(\boldsymbol{X}, t))\, d\boldsymbol{X}.$$

4. Move the structure with $\chi_t = \boldsymbol{u} \circ \chi$.

> Spatial discretization replaces δ with δ_h and evaluates the interaction integrals by quadrature.

FINITE-ELEMENT MOTION AND FORCE FIELDS

With basis functions $\{\phi_\ell\}_{\ell=1}^{M}$, approximate the motion and unified force by

$$\chi_h(\boldsymbol{X},t)=\sum_{\ell=1}^{M}\chi_\ell(t)\phi_\ell(\boldsymbol{X}),\qquad \boldsymbol{\mathcal{F}}_h(\boldsymbol{X},t)=\sum_{\ell=1}^{M}\boldsymbol{\mathcal{F}}_\ell(t)\phi_\ell(\boldsymbol{X}).$$

$$\boldsymbol{F}^h=\nabla_X\chi_h,\qquad \boldsymbol{P}_h=\boldsymbol{P}(\boldsymbol{F}^h).$$

- χ_h is continuous for standard C^0 elements.
- $\boldsymbol{F}^h$ and $\boldsymbol{P}_h$ are generally discontinuous across element boundaries.
- The weak formulation requires elementwise first derivatives, not a pointwise $\nabla_X\cdot\boldsymbol{P}_h$.

> The coefficients χ_ℓ carry geometry. The coefficients $\boldsymbol{\mathcal{F}}_\ell$ represent the projected unified Lagrangian force.

MASS-MATRIX PROJECTION OF THE WEAK FORCE

Testing the unified weak definition with $\boldsymbol{V} = \phi_m \boldsymbol{e}_i$ yields

$$\sum_{\ell=1}^{M} M_{m\ell} \boldsymbol{\mathcal{F}}_\ell = \boldsymbol{B}_m, \qquad M_{m\ell} = \int_{B_0} \phi_m \phi_\ell \, d\boldsymbol{X},$$

$$\boldsymbol{B}_m = -\int_{B_0} \boldsymbol{P}_h \nabla_X \phi_m \, d\boldsymbol{X} + \int_{B_0} \boldsymbol{G} \phi_m \, d\boldsymbol{X}.$$

In block-vector form,

$$M \boldsymbol{c}_{\mathcal{F}} = \boldsymbol{B}, \qquad \boldsymbol{c}_{\mathcal{F}} = \begin{bmatrix} \boldsymbol{\mathcal{F}}_1 & \cdots & \boldsymbol{\mathcal{F}}_M \end{bmatrix}^{\mathsf{T}}.$$

> This L^2 projection converts virtual work into the nodal coefficient vector used by the coupling operators.

Griffith and Luo (2017), Eqs. (25)–(29). The i-th component of $\boldsymbol{B}_m$ is $-\int_{B_0} P_{iI} \, \partial \phi_m / \partial X_I \, d\boldsymbol{X} + \int_{B_0} G_i \phi_m \, d\boldsymbol{X}$; a lumped mass matrix gives a different (but still consistent) discrete inner product.

QUADRATURE-POINT FORCE SPREADING

Let $\boldsymbol{X}_Q^e$ and w_Q^e be quadrature points and weights in element e. The discrete force-spreading operator has the form

$$(\boldsymbol{f}_h)_i = \sum_e \sum_Q \boldsymbol{\mathcal{F}}_h(\boldsymbol{X}_Q^e)\, \delta_h\big(\boldsymbol{x}_i - \chi_h(\boldsymbol{X}_Q^e)\big)\, w_Q^e = (\mathcal{S}_h \boldsymbol{\mathcal{F}}_h)_i.$$

- The FE nodes determine χ_h and the stress approximation.
- Interaction values are evaluated at quadrature points inside each element.
- Adaptive quadrature can keep the interaction net dense as the solid deforms.

> The FE mesh can therefore be coarser than the Eulerian grid without leaving gaps in the coupling stencil, provided the quadrature-point net is sufficiently dense.

Griffith and Luo (2017), Eqs. (31)–(33) and Fig. 2.

ADJOINT VELOCITY RESTRICTION

Let M be the Lagrangian mass matrix and H the Eulerian grid inner-product matrix. Define the nodal velocity by

$$\boxed{\dot{\boldsymbol{\chi}} = \mathcal{R}_h \boldsymbol{u}_h = M^{-1} \mathcal{S}_h^{\mathsf{T}} H \boldsymbol{u}_h}.$$

Let $\boldsymbol{c}_{\mathcal{F}}$ denote the block vector of Lagrangian force coefficients. Then the discrete power identity holds:

$$\underbrace{\boldsymbol{c}_{\mathcal{F}}^{\mathsf{T}} M \dot{\boldsymbol{\chi}}}_{\text{Lagrangian force power}} = \underbrace{(\mathcal{S}_h \boldsymbol{c}_{\mathcal{F}})^{\mathsf{T}} H \boldsymbol{u}_h}_{\text{Eulerian force power}} .$$

> Force spreading and velocity restriction are adjoint under their discrete inner products. This prevents the coupling step from creating or destroying energy by itself.

Griffith and Luo (2017), Eqs. (35)–(40); Peskin (2002). On a uniform 2D grid $H = h^2 I$, which recovers their $[\mathcal{R}] = [M]^{-1} [\mathcal{S}]^{\mathsf{T}} h^2$.

INCOMPRESSIBILITY OF THE IMMERSED MATERIAL

Because $\chi_t = \boldsymbol{u} \circ \chi$, Jacobi's formula gives

$$\frac{\partial J}{\partial t} = J \operatorname{tr}\left(\boldsymbol{F}^{-1} \frac{\partial \boldsymbol{F}}{\partial t} \right) = J \left(\nabla_x \cdot \boldsymbol{u} \right) \circ \chi = 0.$$

Therefore

$$J(\boldsymbol{X}, t) = J(\boldsymbol{X}, 0), \qquad J \equiv 1 \text{ only when } J(\boldsymbol{X}, 0) = 1.$$

At the continuum level, no slip and an incompressible Eulerian velocity preserve solid volume.

At the discrete level, monitor J_h and use an appropriate volumetric stabilization when needed.

> In d dimensions, a common stabilized construction uses $\overline{\boldsymbol{F}} = J^{-1/d} \boldsymbol{F}$ for the isochoric response plus a separate penalty $U(J)$.

Griffith and Patankar (2020), Sec. 2.2.2.

PASSIVE AND ACTIVE STRESS

Biological solids may generate active stress in addition to passive elastic resistance:

$$\boxed{\boldsymbol{P} = \boldsymbol{P}^p + \boldsymbol{P}^a}, \qquad \boldsymbol{P}^p = \frac{\partial W}{\partial \boldsymbol{F}}.$$

The active term $\boldsymbol{P}^a$ may depend on fiber direction, an activation variable, and time. It need not derive from a stored-energy density.

Passive response
Stores and releases elastic energy through $W(\boldsymbol{F})$.

Active response
Supplies mechanical power through prescribed or regulated contraction.

> Hereafter $\boldsymbol{P}$ denotes the total structural PK1 stress. Setting $\boldsymbol{P}^a = \boldsymbol{0}$ recovers the passive convention used in the solid-mechanics module.

ACTIVE STRESS IN THE WEAK-FORCE ASSEMBLY

Substituting $\boldsymbol{P} = \boldsymbol{P}^p + \boldsymbol{P}^a$ into the unified weak form gives

$$\int_{B_0} \boldsymbol{\mathcal{F}} \cdot \boldsymbol{V}\, d\boldsymbol{X} = -\int_{B_0} (\boldsymbol{P}^p + \boldsymbol{P}^a) : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} + \int_{B_0} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{X}.$$

At each structural quadrature point:

1. evaluate $\boldsymbol{P}^p$ from the constitutive law;
2. evaluate $\boldsymbol{P}^a$ from the activation model;
3. add the stresses before assembling the force projection.

> Once expressed as PK1 stress, passive and active contributions use the same FE assembly and fluid-coupling operators.

FIBER-ORIENTED ACTIVE STRESS

Let $\boldsymbol{f}_0(\boldsymbol{X})$ be a unit reference fiber direction and define

$$\boldsymbol{a} = \boldsymbol{F}\boldsymbol{f}_0, \qquad \lambda_f = \|\boldsymbol{a}\|.$$

One unnormalized mapped-fiber convention is

$$\boldsymbol{\sigma}^a = T\,\boldsymbol{a} \otimes \boldsymbol{a},$$

which pulls back to

$$\boxed{\boldsymbol{P}^a = J\boldsymbol{\sigma}^a\boldsymbol{F}^{-\mathsf{T}} = JT\,\boldsymbol{a} \otimes \boldsymbol{f}_0 = JT\,\boldsymbol{F}\boldsymbol{f}_0\boldsymbol{f}_0^{\mathsf{T}}}.$$

> This convention uses the mapped fiber $\boldsymbol{a}$, not its unit direction. A model based on $\boldsymbol{m} = \boldsymbol{a}/\lambda_f$ contains different powers of λ_f; the scalar T must be calibrated for the convention used.

Explicitly: $\boldsymbol{\sigma}^a = T\lambda_f^2\,\boldsymbol{m} \otimes \boldsymbol{m}$, so the Cauchy fiber stress is $T\lambda_f^2$; the unit-direction law $\boldsymbol{\sigma}^a = \widehat{T}\,\boldsymbol{m} \otimes \boldsymbol{m}$ gives $\boldsymbol{P}^a = J\widehat{T}\lambda_f^{-2}\,\boldsymbol{a} \otimes \boldsymbol{f}_0$. The pull-back uses $(\boldsymbol{a} \otimes \boldsymbol{a})\boldsymbol{F}^{-\mathsf{T}} = \boldsymbol{a} \otimes (\boldsymbol{F}^{-1}\boldsymbol{a}) = \boldsymbol{a} \otimes \boldsymbol{f}_0$. The scalar T is unrelated to the bold transmission force $\boldsymbol{T} = -\boldsymbol{P}\boldsymbol{N}$.

JELLYFISH CIRCUMFERENTIAL REFERENCE FIBERS

The subumbrellar swimming musculature is represented by the reference direction

$$\boldsymbol{f}_0(\boldsymbol{X}) = \left(-\frac{X_2}{R},\, \frac{X_1}{R},\, 0 \right), \qquad R = \sqrt{X_1^2 + X_2^2} > 0,$$

where X_1, X_2, X_3 are the scalar components of the reference point $\boldsymbol{X}$ and the bell axis is X_3. With the mapped-fiber convention, the active PK1 stress is

$$\boldsymbol{P}^a = JT(\boldsymbol{X}, t)\, \boldsymbol{F} \boldsymbol{f}_0 \boldsymbol{f}_0^{\mathsf{T}}.$$

- The direction is tangent to circles centered on the bell axis: $|\boldsymbol{f}_0| = 1$ and $\boldsymbol{f}_0 \cdot (X_1, X_2, 0) = 0$.
- Positive activation produces circumferential contractile stress.
- At $R = 0$, the circumferential direction is undefined. Avoid axis quadrature points, regularize the field, or set the activation to zero there.

Active-stress model: convention and activation form as used in Hoover, Griffith & Miller (2017); check against that paper before reusing the constants.

JELLYFISH ACTIVATION FIELD

Write the tension magnitude as

$$T(\boldsymbol{X}, t) = T_{\max}\, \gamma(\boldsymbol{X}, t), \qquad \gamma(\boldsymbol{X}, t) = \alpha(X_3)\, \beta(t),$$

$$0 \le \alpha, \beta, \gamma \le 1.$$

Spatial profile $\alpha(X_3)$
Selects the active band of the bell. A smooth transition avoids a discontinuous stress field.

Temporal profile $\beta(t)$
Controls contraction and relaxation over the pulse cycle. Smooth ramps reduce impulsive forcing.

> Report $T_{\max}$, the spatial and temporal profiles, and the fiber-stress convention together. A value calibrated for a unit-current-fiber convention cannot be transferred unchanged.

Active-stress example: Hoover, Griffith, and Miller (2017). Convention and activation form as used in that paper; check against it before reusing the constants.

LOCALIZED TETHER AS AN EXTERNAL BODY FORCE

An optional restoring force can anchor a selected region:

$$\boldsymbol{G}(\boldsymbol{X},t)=\begin{cases}\kappa\,[\chi(\boldsymbol{X},0)-\chi(\boldsymbol{X},t)], & R\le R_0,\\ \boldsymbol{0}, & R>R_0.\end{cases}$$

- $\boldsymbol{G}$ is a force per reference volume.
- It enters the unified weak form through $\int_{B_0}\boldsymbol{G}\cdot\boldsymbol{V}\,d\boldsymbol{X}$.
- κ has units of force per reference volume per displacement.

> A large tether stiffness can dominate the explicit structural time-step restriction. Verify that the tether does not alter the intended deformation outside the anchored region.

HOMOGENEOUS-STRETCH PROBLEM AND TEST TYPES

For a block $B_0 = [0, L] \times [0, H] \times [0, W]$, prescribe

$$\chi(\boldsymbol{X}) = (\lambda X_1,\, \lambda_\perp X_2,\, \lambda_\perp X_3)\,.$$

- $\lambda_\perp = 1$: uniaxial stretch (prescribed axial deformation);
- $\lambda_\perp = \lambda^{-1/2}$: prescribed isochoric stretch;
- traction-free lateral faces: solve for $\lambda_\perp$ from the constitutive law.

> Prescribing the lateral stretch and prescribing zero lateral traction are different experiments.

HOMOGENEOUS-STRETCH KINEMATICS

For the prescribed motion,

$$\boldsymbol{F} = \nabla_X \chi = \mathrm{diag}(\lambda, \lambda_\perp, \lambda_\perp), \qquad J = \lambda \lambda_\perp^2 .$$

Line and volume elements transform according to

$$d\boldsymbol{x} = \boldsymbol{F}\, d\boldsymbol{X}, \qquad dv = J\, dV.$$

> The motion preserves orientation only for $\lambda > 0$ and $\lambda_\perp > 0$. The special choice $\lambda_\perp = \lambda^{-1/2}$ gives $J = 1$.

STRAIN MEASURES AND INVARIANTS FOR THE STRETCH

$$\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F} = \operatorname{diag}(\lambda^2, \lambda_\perp^2, \lambda_\perp^2),$$
$$\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \tfrac{1}{2}\operatorname{diag}(\lambda^2 - 1, \lambda_\perp^2 - 1, \lambda_\perp^2 - 1).$$

The invariants needed below are

$$I_1 = \lambda^2 + 2\lambda_\perp^2, \qquad I_3 = \det \boldsymbol{C} = J^2 = \lambda^2\lambda_\perp^4.$$

COMPRESSIBLE NEO-HOOKEAN ENERGY AND PARAMETERS

Consider the energy per reference volume

$$W(\boldsymbol{F}) = \frac{\mu_s}{2}(I_1 - 3) - \mu_s \ln J + \frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2, \qquad I_1 = \mathrm{tr}(\boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}).$$

μ_s is the infinitesimal shear modulus.

λ_{L} is the first Lamé parameter, not the bulk modulus.

> In the small-strain limit, the bulk modulus is $K = \lambda_{\mathrm{L}} + 2\mu_s/3$.

Same model and symbols as deck 4-2 (there μ without subscript); compressible neo-Hookean form as in Bonet & Wood (2008). The subscript s distinguishes the solid shear modulus μ_s from the fluid viscosity μ.

DIFFERENTIATING THE NEO-HOOKEAN ENERGY

Use

$$\frac{\partial I_1}{\partial \boldsymbol{F}} = 2\boldsymbol{F}, \qquad \frac{\partial \ln J}{\partial \boldsymbol{F}} = \boldsymbol{F}^{-\mathsf{T}}.$$

Then

$$\boxed{\boldsymbol{P} = \mu_s \left(\boldsymbol{F} - \boldsymbol{F}^{-\mathsf{T}} \right) + \lambda_{\mathrm{L}} (\ln J) \boldsymbol{F}^{-\mathsf{T}}}.$$

> At $\boldsymbol{F} = \boldsymbol{I}$, $J = 1$ and $\boldsymbol{P} = \boldsymbol{0}$, so the reference configuration is stress free.

STRESS RESPONSE OF THE HOMOGENEOUS STRETCH

For $\boldsymbol{F} = \mathrm{diag}(\lambda, \lambda_\perp, \lambda_\perp)$,

$$P_{11} = \mu_s\left(\lambda - \frac{1}{\lambda}\right) + \lambda_{\mathrm{L}}\frac{\ln J}{\lambda},$$

$$P_{22} = P_{33} = \mu_s\left(\lambda_\perp - \frac{1}{\lambda_\perp}\right) + \lambda_{\mathrm{L}}\frac{\ln J}{\lambda_\perp}, \qquad J = \lambda\lambda_\perp^2.$$

- If $\lambda_\perp = 1$, then $P_{22} = P_{33} = \lambda_{\mathrm{L}} \ln \lambda$, generally nonzero.
- A uniaxial-stress test requires $P_{22} = P_{33} = 0$ and determines $\lambda_\perp$: multiplying by $\lambda_\perp$, $\mu_s(\lambda_\perp^2 - 1) + \lambda_{\mathrm{L}} \ln(\lambda\lambda_\perp^2) = 0$.
- An incompressibility constraint $J = 1$ introduces an undetermined pressure and is a different constitutive problem.

UNIFORM STRESS ACTS THROUGH THE BOUNDARY

For the homogeneous test, $\boldsymbol{P}$ is spatially constant, so

$$\nabla_X \cdot \boldsymbol{P} = \boldsymbol{0} \quad \text{inside } B_0.$$

The unified virtual-work term nevertheless remains nonzero:

$$-\int_{B_0} \boldsymbol{P} : \nabla_X \boldsymbol{V}\, d\boldsymbol{X} = -\int_{\partial B_0} (\boldsymbol{P}\boldsymbol{N}) \cdot \boldsymbol{V}\, dA.$$

In the partitioned description,

$$\boldsymbol{\mathcal{F}}_{\text{vol}} = \boldsymbol{0}, \qquad \boldsymbol{T} = -\boldsymbol{P}\boldsymbol{N}.$$

> Constant stress does not mean zero structural force. It means zero interior divergence, with the load carried by boundary tractions.

KINEMATIC TEST AND FULLY COUPLED IBFE SIMULATION

Kinematic constitutive test

- prescribe χ;
- compute $\boldsymbol{F}, J, \boldsymbol{P}$;
- compare against closed-form values;
- use the result to verify material routines.

Fully coupled simulation

- assemble and spread $\boldsymbol{\mathcal{F}}$;
- solve for $\boldsymbol{u}$ and p;
- restrict $\boldsymbol{u}$ to update χ;
- repeat in time.

> The homogeneous stretch validates the solid-side formulas. It is not itself a dynamically coupled IBFE solution.

IBFE VERIFICATION CHECKLIST

Solid mechanics

- verify $J > 0$ and monitor volume drift;
- test objectivity with a rigid rotation;
- compare $\boldsymbol{P}$ with a closed-form homogeneous deformation;
- check boundary tractions and sign conventions.

Eulerian coupling

- verify total-force consistency;
- test the discrete power identity;
- refine the grid, FE mesh, quadrature net, and time step separately;
- check sensitivity to the kernel and volumetric stabilization.

> Constitutive accuracy, FE assembly, and Lagrangian–Eulerian coupling require separate tests.

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Figure credit: IBFE-sketch (configurations slide), schematic from the instructor's course notes.

IBAMR documentation and source: ibamr.github.io